

Microsoft releases C language extension powered by open source

Microsoft has released a new research project called Checked C that is an extension of the existing C language. The new offering emerges as an open source solution to detect and prevent common programming errors.

Unlike developing an all-new language for the developer world, Microsoft Research has modified the original C to offer programmers some new pointer and array types through its Checked C extension. The project itself is written in C and is available under the MIT License.

The availability of new pointer types ensures safety in code. It is aimed at reducing the number of vulnerabilities and puts a stop to the bugs that caused some serious issues like Heartbleed, in the recent past.

“Checked C allows programmers to better describe how they intend to use pointers and the range of memory occupied by data that a pointer points to. This information is then used to add checking at runtime to detect mistakes where the wrong data is accessed, instead of the error occurring silently and without detection. This information can also be used to detect programming errors while the program is being written,” a member of the Microsoft research team wrote in a blog post.



The process of checking is not new and was already debuted on programming languages like C# and Java. But Microsoft's Checked C stretches this 'bounds-checking' by letting developers control the placement of information needed and how the information flows within the code. It helps in providing better control of the program.

Programs that are written in C language need to be converted either file by file or function by function. This enables the detection of errors and offers backward compatibility with existing C developments.

Microsoft has created native Checked C clang and LLVM repositories to provide compiler implementation. The project is additionally accessible through GitHub. However, the project is yet to influence developers in a big way.

Qt 5.7 released with Raspberry Pi 3 support

The Qt Company has released Qt 5.7 as the newest cross-platform application framework. The new version brings native support for Raspberry Pi 3 and uses some QML-based Quick Controls to enhance the user experience.

“I'm very happy to announce that Qt 5.7 is now available,” said Lars Knoll, CTO at the Qt



Company, in a blog post. “It's been only three months since we released Qt 5.6, so one might expect a rather small release with Qt 5.7. But apart from the usual bug fixes and performance improvements, we

have managed to add a whole bunch of new things to this release,” Knoll added.

Qt has always been one of the popular dev tools as over one million developers worldwide are using its features to design multi-platform apps. But the new support for Raspberry Pi 3 is quite likely to increase its existing user base. The community for Raspberry Pi can leverage the new framework and kick-start their embedded development by just installing its SDK.

In addition to Raspberry Pi 3 support, the new Qt version is compatible with advanced hardware such as the NVIDIA Drive CX boards and i.MX7-based devices. Users working on Microsoft Windows can now also use the dev tool as a host platform to develop new offerings for embedded Linux.

Unlike the previous versions, Qt 5.7 supports a C++ 11 compliant compiler to offer developers more of the modern C++ language. It not only adds the functionality of the framework at the root level but also enhances the existing code base of the company with some new features.

“Starting with Qt 5.7, we will require a C++11 compliant compiler to build and use Qt. This allows us to use many of the new features in Qt itself, and renew our code base using features such as constexpr, move semantics, range-for and auto,” Knoll said.

The Qt Company has shifted its licensing from LGPL version 2.1 to LGPL version 3 and GPL version 3 for most of its frameworks and libraries. To promote community efforts, the Finland-based company has additionally open sourced some of its Qt add-ons and tools under GPL version 3.